

王锦改 2024.7.223

第一章作业:

1-10:

解:

标准与式:

$$F = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BCD + \bar{A}B\bar{C}D + A\bar{B}C\bar{D} + ABC\bar{D}$$

标准与式:

$$F = (A+B+C+D)(A+B+\bar{C}+\bar{D})(A+\bar{B}+C+\bar{D})(A+\bar{B}+\bar{C}+D)(\bar{A}+B+C+\bar{D})(\bar{A}+B+\bar{C}+D)(\bar{A}+\bar{B}+C+D)(\bar{A}+\bar{B}+\bar{C}+D)$$

1-12 (7)

$$\begin{aligned} \text{解: } F_2(A, B, C) &= \sum m(0, 1, 4, 6) \\ &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + AB\bar{C} \end{aligned}$$

由最大与项性质约

$$F_2(A, B, C) = \prod M(2, 3, 5, 7)$$

1-14 (6)

$$\begin{aligned} \text{解: } & \bar{A}\bar{B} + (A+C)(C+D) + \overline{(A+B)} + (\bar{C}+\bar{D})E \\ &= \bar{A}\bar{B} + (A+C)(C+D) + \bar{A}\bar{B} + (\bar{C}+\bar{D})E \\ &= \bar{A}\bar{B} + AD + A\bar{B} + CD + \bar{A}\bar{B}C + \bar{C}E + \bar{D}E \\ &= \bar{A}\bar{B} + AD + CD + \bar{A}\bar{B}C + \bar{C}E + \bar{D}E \\ &= \bar{A}\bar{B} + AD + \bar{A}\bar{B}C + CD + \bar{C}DE \\ &= \bar{A}\bar{B} + AD + CD + \bar{B}C + E \end{aligned}$$

1-11 (4)

$$\begin{aligned} \text{解: } F_1 &= A + B\bar{C} + \bar{A}\bar{B}\bar{C} \\ &= A + (B + \bar{A}\bar{B})\bar{C} \\ &= A + B\bar{C} \\ &= A(B+C)(C+\bar{C}) + B\bar{C}(A+\bar{A}) + \bar{A}\bar{B}\bar{C} \\ &= AB(C+\bar{C}) + A\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C} \\ &= \sum m(0, 2, 4, 5, 6, 7) \end{aligned}$$

1-15 (8)

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	0
11	1	1	1	1
10	1	1	1	1

$$\begin{aligned} & ACD + \bar{A}B\bar{C}\bar{D} + \bar{A}\bar{B}(C+\bar{D}) + D \\ &= A + \bar{B} + \bar{C} + D \end{aligned}$$

1-16 d)

解: 卡诺图:

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	1	0	0
11	0	0	1	1
10	0	1	0	1

$$F_1(A, B, C, D) = \bar{A}\bar{C} + \bar{A}\bar{B}D + \bar{B}\bar{C}D + AB\bar{C} + AC\bar{D}$$

1-18 c3)

解: $BC \neq CD = C(B+D) = 0$

AB \ CD	00	01	11	10
00	0	0	1	0
01	0	1	1	0
11	1	1	0	0
10	0	0	1	0

$$F_3 = B\bar{D} + \bar{A}D + \bar{B}D$$

1-17 c5)

解: 卡诺图:

AB \ CD	00	01	11	10
00	1	0	1	0
01	1	0	0	0
11	1	0	1	0
10	0	0	1	0

$$F_5(A, B, C, D) = (C\bar{C} + D)(A + C + \bar{D})(\bar{C}\bar{B} + C + \bar{D})(A + \bar{B} + \bar{D})$$